Karan Choudhary

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Full Cycle AI/ML Engineer (**4+ Years**): Experienced in end-to-end AI product development utilizing a hybrid approach of classic ML (Computer Vision, NLP) and s**tate of the art frameworks (LLM's/Agentic)**. Coming from **agricultural background**, I have keen interest in building solutions focused on high impact areas like **crop monitoring, disease detection, precision farming, and water management.**

EDUCATION

MTech: Indian Institute of Technology, Ropar (Manufacturing Engineering) - 7.9 CGPA **July-2019-Aug-2021**

SKILLS

Generative AI, NLP, Deep Learning, Machine Learning, Computer Vision, Python, Flask, PyTorch, CPS Design, RDF&LPG Databases, Agentic Framework, LLM Use Case Prioritization, Docker

WORK EXPERIENCE

Research & Development Engineer II at Synopsys — Oct-2024-Present

- By exploring, evaluating, and **integrating cutting-edge research, patents, and state of the art tools,** built a system synthesis workflow that formalize and structures ambiguous stakeholder requirements, generate novel system designs and analyzes complex cyber-physical systems.
- Spearheaded a robust **RDF (Ontology) and LPG (Neo4j)** database pipeline using a combination of **LLM, RAG and advanced NLP,** to enable scalable semantic traversal and efficient knowledge retrieval from system data.
- **Developed and deployed end-to-end AI Agentic workflow** leveraging **Computer Vision** and scripting, to autonomously perform trade studies, significantly automating a critical phase of the synthesis workflow.
- Designed and implemented a scalable and reliable **interactive web interface** tool for system synthesis using Flask and Gradio.

Research & Development Engineer at Ubisoft Entertainment — Nov-2022-Oct-2024

- Developed and deployed an in-house algorithm using **OpenCV** to **detect spatial patterns** that cause photosensitive epilepsy, eliminating the need for expensive third-party software, addressing 60% of total visual defects.
- Boosted software testing efficiency nearly 50% by implementing a customized **LLM with RAG** for automated test case generation.
- Developed a proof of concept (POC) desktop application using Streamlit, enabling users to analyze and visualize spatial patterns in videos and images.

Machine Learning Engineer at SLCM Pvt. Ltd. — Nov-2021-Nov-2022

- Led a **team of 5 members**, automating the quality assurance process for a commodity by 80% through the development of a **mobile-based AI/ML application for defective grain detection**.
- Contributed to ML model development by performing **object detection and classification**, curating extensive image datasets, and translating business requirements into actionable model insights.
- Analyzed and benchmarked deep learning architectures, including **Mask R-CNN and YOLO V5**, to determine optimal performance based on time efficiency and detection accuracy for object detection tasks.

PROJECTS

Prediction of laser bending angle – M. Tech Thesis **July-2020-May-2021**

- Laser bending angle was predicted using Adaptive Neuro-fuzzy Inference System (ANFIS)
- Existing data from research was used to train the model with an accuracy of 85%

Generative AI end to end text summarizer

- Text summarizer model was made implementing **RAG** technique on **Llama2** and deployed on **AWS**
- Implemented CI/CD pipeline using **GitHub Actions**